

# KRATANT JAIN

**QA Automation Architect | Lead QE | Test Automation Frameworks | Jenkins CI/CD | Python**  
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## SUMMARY

Technical Specialist with 7+ years across QA automation, CI/CD orchestration, and internal platform engineering. At Sony India Software Centre I built, and now own, the test automation framework, Jenkins orchestration, and device management that run regression for customer-facing SDK releases: ~300 test libraries, ~250,000 test cases, 50+ engineers. Cut regression turnaround from 4–5 days to ~20 hours with 100% on-time milestone delivery. Previously led a 4-member QE team and owned performance testing for a connected-vehicle platform at Harman. Looking for remote Lead QE / SDET Lead or Automation Architect roles.

## KEY RESULTS

- Regression turnaround: 4–5 days → ~20 hours (~75%); ~2 days of manual orchestration prep per cycle eliminated
- CI node environment setup: ~5 hours → ~30 minutes (~90%); devkit setup: ~4 hours, 3 people → ~30 minutes, 1 person (~87%)
- Devkit allocation conflicts: recurring → zero; library onboarding made self-service (~30 libraries added by engineers)
- Milestone delivery: 100% on time, one major SDK milestone delivered early

## EXPERIENCE

### Technical Specialist

Jul 2024 – Present

*Sony India Software Centre, Bengaluru*

Own the automation platform that regression-tests Sony's customer-facing SDK: ~300 test libraries (~250,000 test cases) maintained by 50+ engineers, run across ~8 execution PCs and ~40 devkits. Most of the platform is my own design and code.

- Built the next-generation test automation framework single-handedly, from scratch, as a pip-installable Python package replacing the legacy framework: auto-generated driver scripts, Jenkins execution pathways, and a result-validation utility that catches silent failures.
- Made library onboarding self-service through integration guidelines; ~30 libraries added by engineers from my team and other-region teams without going through the automation team.
- Designed a 3-tier Jenkins orchestration model (super orchestrator → per-PC orchestrator → single-library executor) with up to 10 parallel slots per machine; a spreadsheet-driven run plan removed ~2 days of manual prep per cycle.
- Built a device management service (Flask REST API + NiceGUI dashboard) to reserve, release, and health-check devkits; replaced a pool tool that assigned duplicate IPs, reaching zero allocation conflicts. Now used by other teams.
- Wrote a Python environment-setup framework for Windows CI agents (SDK source detection, zip integrity checks, registry writes with elevated-permission fallback): ~5 hours → ~30 minutes per node; adopted by other regions.
- Automated devkit setup in Jenkins (system/software updates, network configuration): ~4 hours with 3 people → ~30 minutes with 1.
- Built a local CLI mirroring the CI executor so 50+ library maintainers get matching results and reproduce CI failures on any machine; added pre-regression network health monitoring with email alerts and a run-time analytics pipeline.
- Led 50+ engineers through the SVN-to-Git migration using sparse checkout on a 300–400 GB repository; resolved Git LFS pointer-file issues.
- Own delivery: ~5 major regression runs per milestone, go/no-go calls, quality and risk reporting to management, and technical escalation for the full automation stack.
- Drafted the platform AI roadmap: LLM-based failure triage (flaky vs. environment vs. real defect), intelligent reruns, automatic library onboarding from README analysis, self-healing pipelines.

### Senior Test Engineer – Product Development

Apr 2023 – Jul 2024

*Harman Connected Services Corporation India Pvt. Ltd., Bengaluru*

- Led a 4-member QE team for global customers (1 performance, 3 framework maintenance across customers; co-located and remote).
- Owned performance testing for the customer delivery team: weekly JMeter load, stress, and endurance tests with realistic telematics traffic to set server operating limits; traced bottlenecks with Grafana and Graylog.
- Built Jenkins pipelines running performance tests as part of CI; architected a reusable Python + Robot Framework + Selenium framework for web and mobile apps.
- Validated device-to-cloud data flow over MQTT (publish/subscribe, payload integrity, reconnect/retry) for a microservices backend on Kafka, HiveMQ, WSO2, MongoDB, and PostgreSQL.
- Directed regression at each product stage and gave release sign-off; reviewed test cases and scripts; built Python/Pandas data tools and trained developers and QA on automation.

### Product Engineer

Apr 2021 – Mar 2023

*Harman Connected Services Corporation India Pvt. Ltd., Bengaluru*

- Gathered requirements with product and development teams; wrote test plans and detailed test cases for web and mobile telematics applications.
- Ran functional, integration, end-to-end, regression, and exploratory testing; JMeter load, stress, and endurance testing with bottlenecks reported to developers.
- Maintained the known-issues defect database and drove triage and closure; Agile/Scrum releases, code reviews, and mentoring juniors.

#### Programmer Analyst

Aug 2020 – Apr 2021

#### Programmer Analyst Trainee

Jul 2019 – Jul 2020

*Cognizant Technology Solutions*

- Took the Wellington client project live, largely single-handedly, delivering the "one-click fix" (users trigger the fix script on the SCCM server directly instead of raising a ticket); received the PASSION award.
- Automated image-based Citrix workflows with Tesseract OCR + OpenCV where UI automation failed; led to an OCR invoice-reading proof of concept for another client.
- Built a periodic Python ServiceNow user sync replacing manual onboarding; chatbot NLU intent/entity extraction with Rasa Core; Docker deployment of chatbot and platform images; Python MFA proof of concept.

#### Intern, Cloud & Cybersecurity

Jun 2018 – Jul 2018

*Happiest Minds Technologies*

- Built Python tools for SQL injection and XSS checks on given URLs, and network machine discovery with likely vulnerabilities using Nmap.

### SKILLS

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**Languages:** Python, SQL, Groovy (Jenkins DSL), PowerShell, Bash, C

**Automation:** Custom Python frameworks (pip packages), Robot Framework, Pytest, Selenium, Appium/ADB (exposure)

**CI/CD:** Jenkins (multi-tier orchestration, declarative pipelines, conditional stages), Git, Git LFS, SVN

**Performance:** JMeter load/stress/endurance testing, capacity limits, bottleneck analysis

**Web & APIs:** Flask REST APIs, NiceGUI, Django (basics)

**Data & Observability:** Kafka, HiveMQ/MQTT, WSO2, MongoDB, PostgreSQL, Pandas, Grafana, Graylog

**Infrastructure:** Windows CI agents, registry configuration, Docker, Ubuntu/Linux, VMware, AWS (basics)

**Other:** Tesseract OCR, OpenCV, image comparison, NLU/Rasa, SQLi/XSS scanning, Nmap

**Quality Management:** Test strategy, release sign-off, defect governance, JIRA, TestRail, Agile/Scrum

**Learning now:** LangGraph, LangChain, CrewAI, RAG, Chroma, Hugging Face, n8n, MCP, LLM APIs

### EDUCATION

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#### B.Tech, Computer Science Engineering

2015 – 2019

*Jaypee University of Engineering and Technology, Guna*

### CERTIFICATIONS, TRAINING & RECOGNITION

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**Certifications:** Scrum Master; Security White, Yellow and Green Belt; Python Programming Essentials; Python and Flask Bootcamp; Google Digital Unlocked

**Training:** Jenkins DevOps Master; Robot Framework Level 2; Advanced Python; LPIC-2 Linux Kernel and System Startup; AWS Security Fundamentals; CEH v9; CCNA

**Recognition:** PASSION award (Cognizant, Wellington go-live); customer and delivery-manager appreciation

**Languages:** English, Hindi